Classification of Multisource Remote Sensing Data Using Slice Mamba

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***Abstract*— The multisource remote sensing (RS) data have yielded promising results in target detection and classification tasks. However, most existing methods primarily focus on the spatial features inherent in spectral information, while the continuous spectral characteristics are generally neglected. This oversight leads to insufficient extraction of spectral information, thereby limiting detection performance. Recently, the Mamba architecture, based on state space models (SSMs), integrates the advantages of long-range sequence modeling and linear computational efficiency, demonstrating significant potential in low-dimensional scenarios. Inspired by this, we propose Slice Mamba for multisource RS data fusion classification. Specifically, we design two scanning methods: lateral slice scanning (LatSS) and longitudinal slice scanning (LonSS), which construct sequences from lateral and longitudinal perspectives to facilitate information interaction between pixels. In conjunction with the Mamba architecture, we develop the lateral slice Mamba block (LatSMB) and the longitudinal slice Mamba block (LonSMB) to capture continuous spatial–spectral features. Based on this, we establish the slice feature extraction (SFE) module for extracting spatial–spectral feature information and design the cross-information fusion (CIF) module to form a complementary structure for effectively modeling spatial–spectral features, thereby achieving the fusion and classification of multisource heterogeneous features. Experimental results on three benchmark datasets demonstrate that Slice Mamba outperforms existing advanced methods in fusion classification performance and exhibits greater robustness when applied to multispectral datasets.**

***Index Terms*— Mamba, multisource remote sensing (RS) image classification, RS.**

I. Introduction

With the advancement of satellite remote sensing (RS) technology, various types of RS data, such as hyperspectral images (HSIs), multispectral images (MSIs), light detection and ranging (LiDAR), and synthetic aperture radar (SAR), have been generated. These RS data possess distinct characteristics in the classification process. HSI data provide rich spectral information, while the MSI data covers a broader spectral range. LiDAR data offer valuable elevation information, and SAR data excels in the spatial resolution [1], [2], [3], [4]. Although single-source data demonstrates strengths in specific areas, it faces the limitation of balancing multiple advantages, ultimately affecting object classification accuracy [5], [6], [7], [8]. The constraints of single-modal data in classification tasks pose substantial challenges to single-source classification models. However, the diversification of data sources, such as the fusion of spectral data (e.g., HSI or MSI) with LiDAR data, facilitates the formation of complementary relationships between multisource datasets. This approach enables the utilization of the strengths of each data source, leading to more accurate object classification and recognition [9]. The multisource fusion classification integrates these heterogeneous data types, explores potential relationships among them, and overcomes the limitations inherent to single-source data, thereby achieving more effective classification outcomes [5], [10], [11].

Deep learning (DL) effectively achieves the nonlinear feature mapping by adaptively learning parameters through backpropagation and gradient descent algorithms, performing exceptionally well in feature representation. As a result, DL is widely applied in multisource RS image classification [12], [13], [14], [15], [16]. Typically, multisource fusion and classification frameworks based on DL employ a multistream architecture, where each stream corresponds to a distinct data source. Building on this design, Xu et al. [17] proposed a dual-branch convolutional neural networks (CNNs), in which the HSI branch is modified into a dual-tunnel CNNs framework to simultaneously extract spectral and spatial features, fully utilizing the spatial–spectral information in hyperspectral images. Similarly, Hang et al. [18] developed an efficient framework for fusing HSI and LiDAR data using dual-coupled CNNs. In their model, one CNNs captures spatial–spectral features from hyperspectral images, while the other extracts elevation information from LiDAR data. These two networks are coupled through a parameter-sharing strategy, which reduces the overall number of parameters. In addition, Zhang et al. [19] introduced a structured optimized transmission network (SOT-Net), which adaptively adjusts imbalanced distributions, reduces redundant interference between signal

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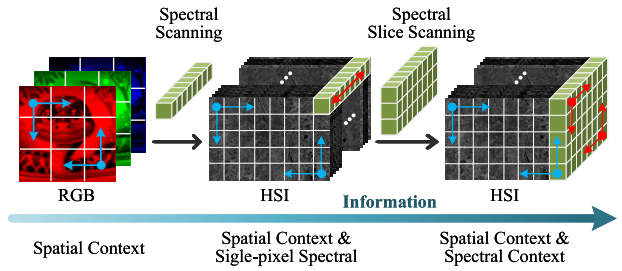


Fig. 1. Motivation of the proposed Slice Mamba. The current Mamba method developed for HSI integrates spectral data. However, it remains insufficient in effectively capturing the relationships between adjacent pixels in the spectral modeling.

sources, and enhances collaborative classification performance. Although CNN-based models demonstrate promising results, they still face limitations in capturing global spectral and spatial characteristics and have difficulty modeling long-range dependencies between pixels.

Benefiting from the powerful global modeling capabilities of attention mechanisms, the application of Transformer architectures in multisource fusion classification tasks has been extensively studied in recent years [20], [21], [22], [23]. Yang et al. [24] proposed a novel cross-attention mechanism using the Transformer architecture, where LiDAR data guide the selection of HSI bands. Zhao et al. [25] explored a plug-and-play image transformer, fractional Fourier image transformer (FrIT), which utilizes multimodal global context and sequential features, integrating them into the ViT model to enhance robustness and generalization. Zhang et al. [26] harness the strengths of transformers to capture both specific and shared characteristics of hyperspectral and LiDAR data, learning spectral local sequence information from adjacent bands of hyperspectral images while globally mining pixel-level feature embedding relationships. They also capture spatial and elevation information separately from the hyperspectral and LiDAR data. Ni et al. [27] proposed a frequency-local feature learning framework based on the discrete wavelet transform (DWT), which includes a three-branch network comprising spectral, spatial, and elevation learning and enhancement modules. This framework conducts spatial feature learning on low-frequency components and adaptive feature learning on high-frequency components. Despite the impressive capabilities of Transformer architectures in HSI classification, the inherent computational complexity of their self-attention mechanisms, which is $O(N^2)$, poses significant challenges in terms of modeling efficiency and memory overhead, particularly in multibranch scenarios [26], [28].

In contrast, recent state space models (SSMs) [29] establish long-range dependencies using the HIPPO matrix, offering excellent attributes, such as linear computational complexity and scalability. As an effective alternative to transformers, Mamba introduces selective SSMs for sequence modeling in specific directions, demonstrating substantial potential in natural language processing (NLP) tasks and applied to the field of medical imaging [30], [31], [32], [33] and hyperspectral RS [34], [35], [36], [37]. To accommodate visual scenarios involving 2-D spatial awareness, Vim [38] and VMamba [39] extend the Mamba architecture by incorporating multidirectional scanning mechanisms to achieve global contextual modeling, showing efficiency and effectiveness in image tasks such as object detection and semantic segmentation. As shown in Fig. 1, natural images are acquired through spatial scanning to obtain structural information for feature extraction. In the field of hyperspectral imaging, the spectral scanning is further developed on the basis of spatial scanning to acquire richer spectral information. Building on the traditional spectral scanning, we propose a slice scanning method that incorporates additional spectral scanning directions. This is necessary because existing scanning methods focus solely on individual pixel spectral channels, resulting in a lack of connectivity between adjacent pixels and leading to the loss of contextual structural information between them [35], [40]. This article successfully extracts spatial–spectral features from HSIs through a slicing approach. Building on this slicing technique, we achieve the fusion classification of multisource data.

Specifically, this article proposes a Slice Mamba framework based on the SSM for efficient global spatial–spectral context modeling in the multisource RS image classification. The Slice Mamba framework includes a multiscale information fusion (MIF) module, a slice feature extraction (SFE) module, a cross-information fusion (CIF) module, and a prediction module. The MIF module utilizes convolutional kernels of varying sizes to extract and fuse shallow features at different scales. The SFE module consists of the lateral Slice Mamba block (LatSMB) and the longitudinal Slice Mamba block (LonSMB), which are designed to extract deep features from the data. To address the issue of traditional Mamba models being unable to capture the similarities between adjacent pixels during spectral modeling, we develop lateral slice scanning (LatSS) and longitudinal slice scanning (LonSS) mechanisms, which serve as core components for LatSMB and LonSMB, respectively. After scanning, pixelwise flattening is performed, followed by selective scanning using the S6 model to facilitate sufficient interaction between adjacent pixels, thereby enabling spatial–spectral sequence modeling. The CIF module adopts a complementary structure to the SFE module, utilizing features extracted with the LatSS scanning mechanism in the SFE module and applying the LonSS scanning mechanism in the CIF module. This cross-scanning approach forms a complementary structure that comprehensively extracts the spatial–spectral features of the data. Finally, the prediction module predicts pixel classes based on the extracted features.

The main contributions are summarized as follows.

1) This article proposes a dual-branch Slice Mamba framework for multisource RS data fusion and classification tasks, which improves classification accuracy for fusion tasks while maintaining linear complexity.
2) We design two scanning mechanisms, LatSS and LonSS, to enhance information interaction between adjacent pixels from a sequence modeling perspective, which are used to construct the LatSMB and LonSMB modules. Combined with the selective scanning mechanism of the Mamba architecture, this approach establishes long-range spatial–spectral contextual dependencies with

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linear complexity, addressing the limitations of existing Mamba architectures in the spectral feature extraction.

3) The SFE and CIF modules are constructed, where the SFE module consists of LatSMB and LonSMB to extract spatial structural and spectral information, respectively. The CIF module adopts a complementary structure to SFE, applying LonSMB to the features generated by LatSMB in SFE, thereby achieving information complementarity, comprehensive feature extraction, and heterogeneous information fusion.
4) Extensive experiments are conducted on three public hyperspectral datasets to evaluate the effectiveness and superiority of the proposed method. Specifically, we use hyperspectral and LiDAR datasets and extend validation to MSI and LiDAR datasets to confirm the robustness of the proposed algorithm.

The method proposed in this article is applicable to multisource fusion classification tasks. The remainder of this article is organized as follows. Section II provides a detailed introduction to the proposed Slice Mamba method; Section III presents the experimental results and analysis; finally, Section IV summarizes the main conclusion.

II. Proposed Network

This section first introduces the preliminary knowledge of SSMs. Subsequently, the Slice Mamba framework for multisource RS image fusion and classification is presented. The MIF module for multisource data is then described. In addition, the LatSS and LonSS mechanisms are introduced, leading to the development of the corresponding LatSMB and LonSMB modules. Finally, the CIF module for multisource data integration is presented.

A. Preliminaries

SSMs: Recently, the advent of structured state space sequence models (S4) and selective state space sequence models (S6) has garnered significant attention in the field of sequence modeling. This modeling approach originates from classical linear time-invariant systems. Given a continuous function $x(t) \in \mathbb{R}$ as input, the response $y(t) \in \mathbb{R}^N$ is determined via the hidden state $h(t) \in \mathbb{R}^N$. The process can be described by the following ordinary differential equation (ODE):

$$\begin{aligned} h(t) &= \mathbf{A}h(t-1) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t) \end{aligned} \tag{1}$$

where the matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ governs the relationship with the previous state of the node, while the matrix $\mathbf{B} \in \mathbb{R}^{N\times 1}$ captures the factors influencing the system's response to the current input. The matrix $\mathbf{C} \in \mathbb{R}^{N\times 1}$ is associated with the hidden state, and $\mathbf{D} \in \mathbb{R}^1$ represents the residual connection, which is omitted in this analysis for the sake of simplification.

Since images are stored in discrete form on computers, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ in (1) must be discretized using the zero-order hold (ZOH) technique to accommodate image-related tasks. This discretization process can be represented as follows:

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\Delta\mathbf{B} \\ &\approx (\Delta\mathbf{A})^{-1}(\Delta\mathbf{A})(\Delta\mathbf{B}), \\ &= \Delta\mathbf{B} \end{aligned} \tag{2}$$

where Δ represents the time step and the parameters $\mathbf{A}$ and $\mathbf{B}$ are discretized as $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. Thus, (1) can be reformulated as follows:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

In the S4 architecture, the parameters are static, thus enabling the linear recurrence process to be efficiently computed through a convolutional form, which can be expressed as follows:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right) \\ y &= x * \overline{\mathbf{K}} \end{aligned} \tag{4}$$

where L is the length of the input sequence $x, \overline{\mathbf{K}} \in \mathbb{R}^L$ can be regarded as a kernel, and $*$ denotes the convolutional operation.

Although the S4 architecture demonstrated strong performance, it exhibited suboptimal results in downstream tasks because its parameters were static and independent of the input [29]. Building on S4, and S6 addresses this limitation by making the parameters dependent on the input data. Specifically, given the input $x \in \mathbb{R}^{B\times L\times D}$, the dynamic parameters $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are determined by the input x, enabling interactions between the input data and the system states.

B. Slice Mamba: Overview

Given a hyperspectral image $\mathbf{X}_H \in \mathbb{R}^{H\times W\times C}$ and a LiDAR image $\mathbf{X}_L \in \mathbb{R}^{H\times W\times 1}$ covering the same area on the Earth's surface, where H and W represent the spatial size and C is the number of HSI spectral bands. HSI contains abundant spectral information, and the large number of spectral bands often leads to significant information redundancy, resulting in the Hughes phenomenon. To mitigate this effect, principal component analysis (PCA) is applied to perform dimensionality reduction on the HSI data. The dimensionality reduced HSI is represented as $\mathbf{X}_H^{\text{PCA}} \in \mathbb{R}^{H\times W\times c}$. Then, a small cube $x_h \in \mathbb{R}^{p\times p\times c}$ and a small patch $x_l \in \mathbb{R}^{p\times p\times 1}$ are selected from $\mathbf{X}_H^{\text{PCA}}$ and $\mathbf{X}_L$, respectively, as inputs to the Slice Mamba.

The flowchart of the proposed Slice Mamba model for multisource RS image fusion and classification is depicted in Fig. 2(a). The Slice Mamba model adopts a dual-branch architecture. Inspired by NCGLF2 [5], the MIF module is employed for single-source data processing, while LatSMB and LonSMB are utilized for deep feature extraction. After the MIF, SFE, and CIF modules, the classification is ultimately achieved.

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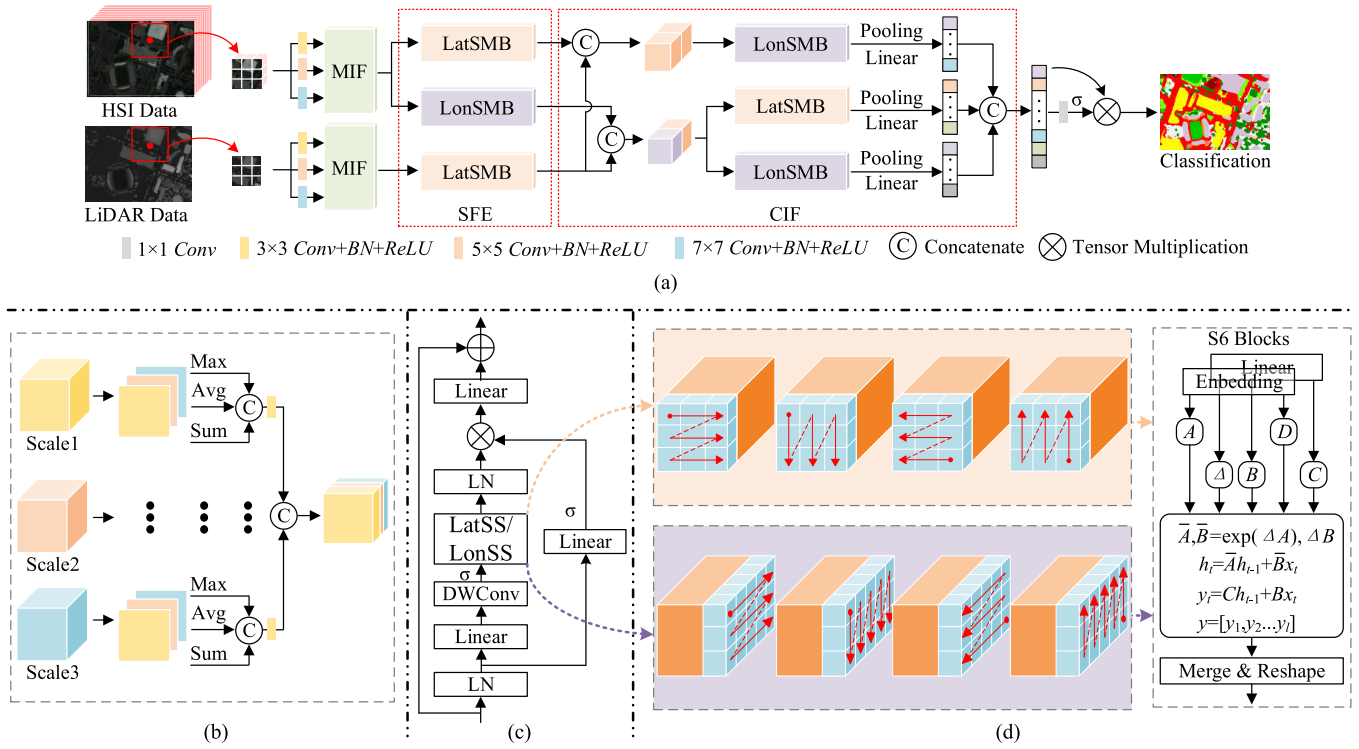


Fig. 2. (a) Overall architecture of Slice Mamba for multisource RS image fusion classification, which includes MIF module, SFE module, and CIF module, SFE module is composed of the LatSMB iterated from layer L_1 and the LonSMB iterated from layer L_2. (b) Structural flow of MIF module. (c) Structural flow of LatSMB and LonSMB. (d) Computational procedure of LatSS and LonSS.

Initially, x_h and x_l are projected into the same dimensional space using distinct convolutional kernels, formally expressed as follows:

$$\begin{aligned} f_h^m &= \varphi_{\mathrm{ReLU}}(\varphi_{\mathrm{BN}}(\varphi_{\mathrm{Conv}_{m\times m}}(x_h))) \\ f_l^m &= \varphi_{\mathrm{ReLU}}(\varphi_{\mathrm{BN}}(\varphi_{\mathrm{Conv}_{m\times m}}(x_l))) \end{aligned} \tag{5}$$

where f_h^m and f_l^m represent the features of x_h and x_l, respectively, obtained through convolutional operations with kernel sizes of $m \times m$, followed by batch normalization (BN) and rectified linear unit (ReLU) activation. Specifically, the values of m are set to 3, 5, and 7.

In MIF module, as illustrated in Fig. 2(b), taking f_h^m as an example, the maximum value $f_h^{\mathrm{Max}}(i)$, sum value $f_h^{\mathrm{Sum}}(i)$, and average value $f_h^{\mathrm{Avg}}(i)$ are obtained by comparing $f_h^3(i)$, $f_h^5(i)$, and $f_h^7(i)$, where i represents the feature dimensions after convolution. The fused features $F_h(i)$ and $F_l(i)$ are then obtained through the concatenation operation, as expressed as follows:

$$F_h(i) = \mathrm{Cat}\left(f_h^{\mathrm{Max}}(i),\, f_h^{\mathrm{Sum}}(i),\, f_h^{\mathrm{Avg}}(i)\right) \tag{6}$$

where $f_h^{\mathrm{Max}}(i)$, $f_h^{\mathrm{Sum}}(i)$, and $f_h^{\mathrm{Avg}}(i)$ can be expressed as follows:

$$\begin{aligned} f_h^{\mathrm{Max}}(i) &= \max\left(f_h^3(i),\, f_h^5(i),\, f_h^7(i)\right) \\ f_h^{\mathrm{Sum}}(i) &= \mathrm{sum}\left(f_h^3(i),\, f_h^5(i),\, f_h^7(i)\right) \\ f_h^{\mathrm{Avg}}(i) &= \mathrm{avg}\left(f_h^3(i),\, f_h^5(i),\, f_h^7(i)\right). \end{aligned} \tag{7}$$

Similarly, $F_l(i)$ can be derived using the same approach.

Subsequently, both F_h and F_l are employed as inputs for SFE and are subsequently fed into the SFE module. In particular, the spatial information, denoted by F_h^{Lat}, and the spectral information, denoted by F_h^{Lon}, are extracted from F_h using LatSMB and LonSMB, respectively, and are expressed as follows:

$$\begin{aligned} F_h^{\mathrm{Lat}} &= \varphi_{\mathrm{LatSMB}}(F_h) \\ F_h^{\mathrm{Lon}} &= \varphi_{\mathrm{LonSMB}}(F_h). \end{aligned} \tag{8}$$

At the same time, LatSMB is used to obtain the spatial information in F_l^{Lon}, which is expressed as follows:

$$F_l^{\mathrm{Lat}} = \varphi_{\mathrm{LatSMB}}(F_l). \tag{9}$$

After that, the feature fusion is performed through the CIF module and can be expressed as follows:

$$F_{\mathrm{fusion}} = \varphi_{\mathrm{CIF}}\left(F_h^{\mathrm{Lat}},\, F_h^{\mathrm{Lon}},\, F_l^{\mathrm{Lat}}\right). \tag{10}$$

Finally, the fused features F_{fusion} are enhanced using a channel attention mechanism, and the enhanced features F can be represented as follows:

$$F = F_{\mathrm{fusion}} \otimes \sigma\left(\varphi_{\mathrm{Conv}_{1\times1}}(F_{\mathrm{fusion}})\right) \tag{11}$$

where σ represents the Sigmoid function and $\otimes$ denotes elementwise multiplication. It is processed by $\varphi_{\mathrm{Classifier}(\cdot)}$ to generate the final classification results

$$y = \varphi_{\mathrm{Classifier}}(F). \tag{12}$$

The classifier consists of one perceptron layer with an embedding dimension of 32, mapping the fused spatial–spectral features into category results. The classification is optimized with the cross-entropy loss function. The training procedure is summarized in Algorithm 1.

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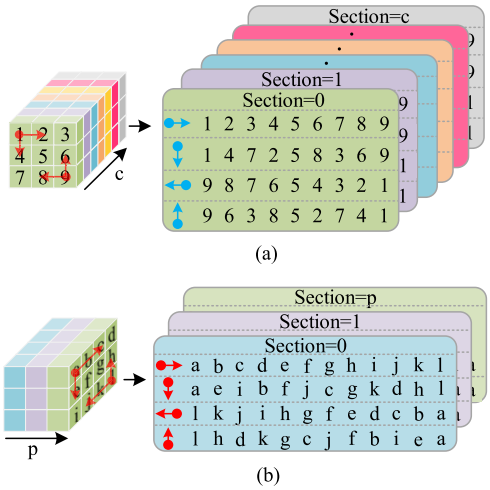


Fig. 3. (a) Scanning process of LatSS. (b) Scanning process of LonSS.

C. Slice Feature Extraction

HSIs contain rich spatial and spectral information, this article effectively extracts this information by stacking the LatSMB in the L_1 layer and the LonSMB in the L_2 layer. Inspired by VMamba [39], we adopt a multidirectional scanning approach to capture features. Unlike VMamba, we develop two distinct scanning methods, LatSS and LonSS, as key steps in LatSMB and LonSMB, tailored specifically for hyperspectral tasks. These methods are illustrated in Fig. 2(c) and (d).

The input F_h is first subjected to layer normalization (LN)

$$f_1 = \varphi_{\mathrm{LN}}(F_h). \tag{13}$$

The input sequentially passes through the linear layer and DWConv layer before entering either LatSS or LonSS and is normalized using LN. The specific scanning processes of LatSS and LonSS are illustrated in Fig. 3

$$\begin{aligned} f_2' &= \sigma(\varphi_{\mathrm{DWConv}_{3\times3}}(\varphi_{\mathrm{Linear}}(f_1))) \\ f_2'' &= \varphi_{\mathrm{LatSS}}/\varphi_{\mathrm{LonSS}}(f_2') \\ f_2 &= \varphi_{\mathrm{LN}}(f_2''). \end{aligned} \tag{14}$$

f_1 passes through the linear layer and sigmoid activation, and then mulitply with f_2

$$f_3 = f_2 \times \sigma(\varphi_{\mathrm{Linear}}(f_1)). \tag{15}$$

f_3 passes through the linear layer and undergoes a skip connection with f_h, resulting in the final output F_h^{Lat}

$$F_h^{\mathrm{Lat}} = f_h + \varphi_{\mathrm{Linear}}(f_3)). \tag{16}$$

1) LatSS: LatSS serves as the core component for LatSMB, constructing spatial structural sequences through lateral scanning. First, f_2' is sliced laterally, as shown in Fig. 3(a), and then flattened sequentially from left to right, top to bottom, right to left, and bottom to top

$$\begin{aligned} f^{\mathrm{LR}} &= \varphi_{\mathrm{flatten}}^{\mathrm{LR}}(f_2', p^2) \\ f^{\mathrm{TB}} &= \varphi_{\mathrm{flatten}}^{\mathrm{TB}}(f_2', p^2) \\ f^{\mathrm{RL}} &= \varphi_{\mathrm{flatten}}^{\mathrm{RL}}(f_2', p^2) \\ f^{\mathrm{BT}} &= \varphi_{\mathrm{flatten}}^{\mathrm{BT}}(f_2', p^2). \end{aligned} \tag{17}$$

Algorithm 1 Pseudoprocedure of the Proposed Slice Mamba

Input: HSI image $X_H \in \mathbb{R}^{H\times W\times C}$, LiDAR image $X_L \in \mathbb{R}^{H\times W\times 1}$, depths L_1 and L_2.
Output: Predicted labels of the testing samples.

```
Apply PCA for dimension reduction X_H^{PCA} ∈ R^{H×W×c}.
Crop patch x_h ∈ R^{p×p×c} and x_l ∈ R^{p×p×1}, with labels determined by the central pixel.
MIF module:
Perform single-source data processing on x_h and x_l based on Eq. (5), generating shallow multiscale features F_h and F_l.
SFE module:
for depth i = 1 to L_1 do
    Perform lateral slice modeling on F_h and F_l based on Eq. (8) and Eq. (9), generating HSI spatial feature F_h^{Lat} and LiDAR spatial feature F_l^{Lat}, respectively.
end for
for depth j = 1 to L_2 do
    Perform longitudinal slice modeling on F_h based on Eq. (8), generating HSI spectral feature F_h^{Lon}.
end for
CIF module:
Perform feature fusion based on Eq. (22), generating multimodal spatial-spectral features F_hl^{Lat−Lat} and F_hl^{Lat−Lon}.
for depth i = 1 to L_1 do
    Perform lateral slice modeling on F_hl^{Lat−Lon} based on Eq. (23), generating heterogeneous feature F_2.
end for
for depth j = 1 to L_2 do
    Perform longitudinal slice modeling on F_hl^{Lat−Lat} and F_hl^{Lat−Lon} based on Eq. (23), generating heterogeneous feature F_1 and F_3, respectively.
end for
Perform adaptive feature concatenation based on Eq. (11), generating output F.
Classifier:
Perform ultimate classification based on Eq. (12), generating final results y.
```

Subsequently, these sequences are fed into the S6 module for state space evolution, as shown in Fig. 2(d). Taking $f^{\mathrm{LR}} = \{[x_0, x_1, \ldots, x_{p^2}]\}|x_j \in \mathbb{R}^{1\times c}$ as an example, the process can be formulated as follows:

$$\begin{aligned} h_j &= \overline{\mathbf{A}}_{\mathrm{Lat}} h_{j-1} + \overline{\mathbf{B}}_{\mathrm{Lat}} x_j \\ y_j &= \mathbf{C}_{\mathrm{Lat}} h_j \end{aligned} \tag{18}$$

where $\overline{\mathbf{A}}_{\mathrm{Lat}}$, $\overline{\mathbf{B}}_{\mathrm{Lat}}$, and $\mathbf{C}_{\mathrm{Lat}}$ are the training parameters of the S6 model. After efficient computation, the transformed output can be expressed as $f_{\mathrm{Lat}}^{\mathrm{LR}} = \{[y_0, y_1, \ldots, y_{p^2}]\}|y_j \in \mathbb{R}^{1\times c}$. Notably, the same procedure is executed for the remaining three parallel groups, with different scanning directions and independent training parameters.

After the parallel processing, the generated output sequences are subjected to the concatenation operation

$$f_2'' = \varphi_{\mathrm{cat}}(f_{\mathrm{Lat}}^{\mathrm{LR}}, f_{\mathrm{Lat}}^{\mathrm{TB}}, f_{\mathrm{Lat}}^{\mathrm{RL}}, f_{\mathrm{Lat}}^{\mathrm{BT}}). \tag{19}$$

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TABLE I
CATEGORY NAME, SAMPLE NUMBERS OF TRAINING SET AND TESTING SET OF EACH CLASS ON THE HOUSTON 2013, AUGSBURG, AND GRSS-DFC-2007 DATASETS

Class	Houston 2013			Augsburg			GRSS-DFC-2007		
	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Health grass	20	1231	Forest	344	13163	City Center	674	12811
2	Stressed grass	20	1234	Residential-Area	334	29995	Residential Areas	1481	28136
3	Synthetic grass	20	677	Industrial-Area	112	3739	Sparse Building	752	14282
4	Trees	20	1224	Low-Plants	1162	25695	Water	45	848
5	Soil	20	1222	Allotment	91	484	Vegetation	30	561
6	Water	20	305	Commercial-Area	23	1622			
7	Residential	20	1248	Water	312	1218			
8	Commercial	20	1224						
9	Road	20	1232						
10	Highway	20	1207						
11	Railway	20	1215						
12	Parking Lot 1	20	1213						
13	Parking Lot 2	20	449						
14	Tennis Court	20	408						
15	Running Track	20	640						
	Total	300	14729	Total	2378	75916	Total	2982	56638

2) LonSS: LonSS serves as the core component for LonSMB, constructing spectral structure sequences through longitudinal scanning. The scanning process is consistent with that of LatSS, as shown in (17). Therefore, the S6 process for $f^{\mathrm{LR}} = \{[x_0, x_1, \ldots, x_{p\times c}]\} | x_j \in \mathbb{R}^p$ can be formulated as follows:

$$\begin{aligned} h_j &= \overline{\mathbf{A}}_{\mathrm{Lon}} h_{j-1} + \overline{\mathbf{B}}_{\mathrm{Lon}} x_j \\ y_j &= \mathbf{C}_{\mathrm{Lon}} h_j \end{aligned} \tag{20}$$

where $\overline{\mathbf{A}}_{\mathrm{Lon}}, \overline{\mathbf{B}}_{\mathrm{Lon}}$, and $\mathbf{C}_{\mathrm{Lon}}$ denote the training parameters. Similarly, the output f_2'' can be expressed as follows:

$$f_2'' = \varphi_{\mathrm{cat}}\left(f_{\mathrm{Lon}}^{\mathrm{LR}}, f_{\mathrm{Lon}}^{\mathrm{TB}}, f_{\mathrm{Lon}}^{\mathrm{RL}}, f_{\mathrm{Lon}}^{\mathrm{BT}}\right). \tag{21}$$

D. Cross-Information Fusion

The scanning structures of the CIF and SFE modules complement each other, aiming to fully extract spatial–spectral features. The spatial features of the HSI, extracted via LatSMB, are fused with the spatial features of the LiDAR image through concatenation. These fused features are further integrated using LonSMB. Simultaneously, the spectral features of the HSI, extracted by LonSMB, are combined with the spatial features of the LiDAR image through concatenation

$$\begin{aligned} F_{\mathrm{hl}}^{\mathrm{Lat-Lat}} &= \varphi_{\mathrm{cat}}\left(F_h^{\mathrm{Lat}}, F_l^{\mathrm{Lat}}\right) \\ F_{\mathrm{hl}}^{\mathrm{Lat-Lon}} &= \varphi_{\mathrm{cat}}\left(F_h^{\mathrm{Lon}}, F_l^{\mathrm{Lat}}\right). \end{aligned} \tag{22}$$

These features are then integrated using both LatSMB and LonSMB, forming a complementary structure that facilitates the comprehensive extraction of heterogeneous features. This process can be represented as follows:

$$\begin{aligned} F_1 &= \varphi_{\mathrm{LonSMB}}\left(F_{\mathrm{hl}}^{\mathrm{Lat-Lat}}\right) \\ F_2 &= \varphi_{\mathrm{LatSMB}}\left(F_{\mathrm{hl}}^{\mathrm{Lat-Lon}}\right) \\ F_3 &= \varphi_{\mathrm{LonSMB}}\left(F_{\mathrm{hl}}^{\mathrm{Lat-Lon}}\right) \end{aligned} \tag{23}$$

then, the integrated features are fused through a concatenation mechanism

$$F_{\mathrm{fusion}} = \varphi_{\mathrm{cat}}\{F_1, F_2, F_3\}. \tag{24}$$

III. PERFORMANCE EVALUATION

This section first introduces the three datasets used in this article. Next, it describes the experimental setup, including evaluation metrics, hardware and software configurations, dataset presentation, and the training and testing set configurations. A quantitative analysis of the experimental parameters is then conducted to determine the optimal parameters, followed by a comparative experiment. Subsequently, an analysis of computational complexity is provided, and finally, the necessity of each module is validated through ablation experiments.

A. Datasets Description

To verify the effectiveness of Slice Mamba, we perform extensive experiments on three multisource RS datasets. The details of the datasets used in the experiments are provided below. The data settings of training data and testing data are shown in Table I.

1) Houston 2013: The dataset comprises scenes acquired by the Compact Airborne Spectral Imager (CASI) over the University of Houston campus and its surrounding urban areas [41]. It includes both HSI- and LiDAR-based digital surface model (DSM) data, with a spatial resolution of 2.5 m and dimensions of 349 × 1905 pixels. The HSI data consists of 144 spectral bands, spanning a wavelength range of 0.38–1.35 μm. Furthermore, the dataset is annotated with 15 029 labeled samples, representing 15 distinct land cover categories. Fig. 4 shows a pseudocolor composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map.

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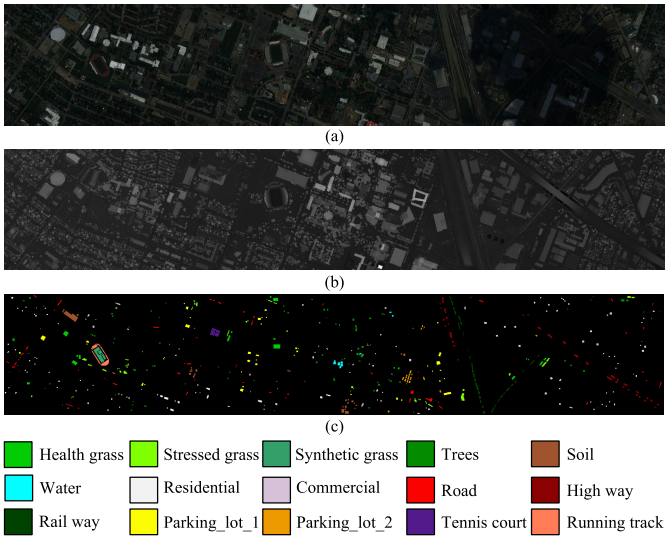


Fig. 4. Visualization of the Houston 2013 data. (a) Pseudocolor image for the HSI data using 64, 32, and 22 as R, G, and B, respectively. (b) Grayscale image for the LiDAR data. (c) Ground-truth map.

2) Augsburg: This dataset consists of HSI data and DSM-based LiDAR data collected from the city of Augsburg, Germany, and its surrounding areas [42]. The HSI data contains 180 spectral bands, with a wavelength range from 0.4 to 2.5 μm, obtained using the DAS-EOC HySpex sensor. The DSM data based on LiDAR is collected using the DLR-3 K system. Both images are set to a spatial resolution of 30 m, with a pixel size of 332 $\times$ 485, covering seven different land cover classes. The ground reference data are generated from OpenStreetMap data. Fig. 5 shows a pseudocolor composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map.

3) GRSS-DFC-2007: The GRSS-DFC-2007 dataset is utilized in the 2007 Data Fusion Contest, covering the geographical region of the urban area of Pavia, located in northern Italy [43]. This dataset includes MSI and SAR imagery. The MSI data are acquired by the Landsat-5 sensor and consist of six spectral bands, while the SAR imagery is obtained from the ERS-1 sensor. Both types of images are provided at a resolution of 787 787 pixels, and the dataset encompasses five distinct land cover classes. Fig. 6 shows a pseudocolor composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map.

B. Experimental Settings

1) Evaluation Metrics: This article adopts three evaluation metrics: overall accuracy (OA), average accuracy (AA), kappa coefficient (kappa), and classwise accuracy. OA measures the classification accuracy of the model across the entire dataset. AA balances the classification performance across different categories, while the Kappa effectively assesses the performance of the classifier in the presence of class imbalance. The values of these metrics range from 0 to 1, and generally, higher values indicate better performance. To ensure a fair comparison, the reported results are the average of ten consecutive experiments.

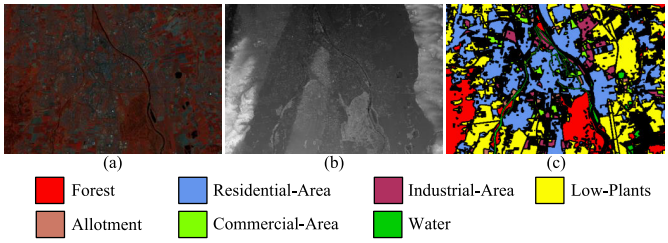


Fig. 5. Visualization of the Augsburg data. (a) Pseudocolor image for the HSI data using 50, 100, and 150 as R, G, and B, respectively. (b) Grayscale image for the LiDAR data. (c) Ground-truth map.

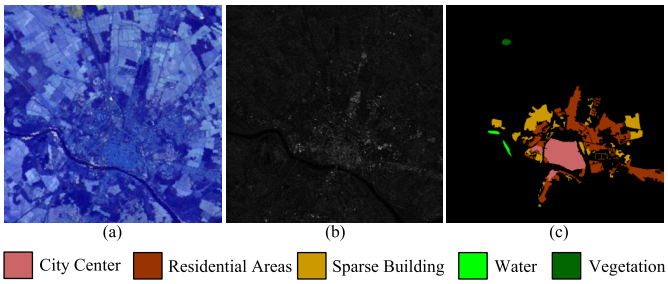


Fig. 6. Visualization of the GRSS-DFC-2007 data. (a) Pseudocolor image for the MSI data using 1, 2, and 3 as R, G, and B, respectively. (b) Grayscale image for the LiDAR data. (c) Ground-truth map.

2) Configurations: We build our Slice Mamba on the basis of the PyTorch machine learning framework using a PC with 64-GB memory and an Nvidia GeForce RTX 4060Ti GPU card. The training epochs and batch size are set as 80 and 64, respectively. The Adam optimizer, with a learning rate set to 0.001, is utilized for optimizing the parameters.

C. Parameter Analysis

In this section, we conduct a series of experiments to analyze and determine the parameters of Slice Mamba, including the input patch size, the number of dimensions after PCA reduction, and the number of layers for LatSMB and LonSMB, denoted by L_1 and L_2.

1) Input Patch Size: Fig. 7 illustrates the impact of different patch sizes on classification performance across various datasets. Patch sizes ranging from 3 $\times$ 3 to 11 $\times$ 11 are considered. For the Houston 2013 dataset, accuracy initially increases with larger patch sizes, reaches its peak at 9 $\times$ 9, and then decreases. In the Augsburg dataset, OA and Kappa increase with larger patch sizes before stabilizing, while AA continues to improve as the patch size increases. In the GRSS-DFC-2007 dataset, accuracy steadily improves with increasing patch sizes. Based on these observations, the proposed method employs a patch size of 9 $\times$ 9 for the Houston 2013 dataset and 11 $\times$ 11 for both the Augsburg and GRSS-DFC-2007 datasets.

2) Dimensionality Reduction via PCA: Fig. 8 illustrates the impact of different dimensionalities, following PCA reduction, on classification performance across various datasets. For the Houston 2013 and Augsburg datasets, we evaluate five dimensions ranging from 1 to 40. The results show that as the number of dimensions increases, classification performance initially improves and then stabilizes. In the Houston

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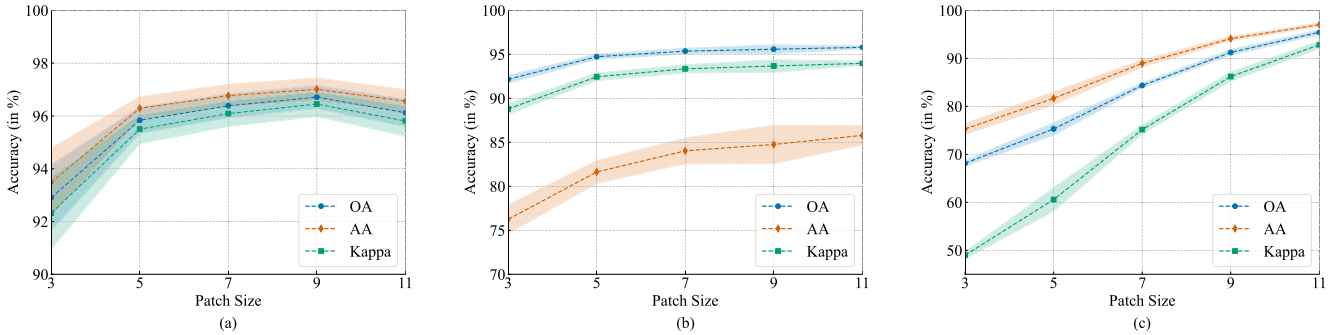


Fig. 7. Impact of different patch sizes on accuracy in (a) Houston 2013 dataset, (b) Augsburg dataset, and (c) GRSS-DFC-2007 dataset.

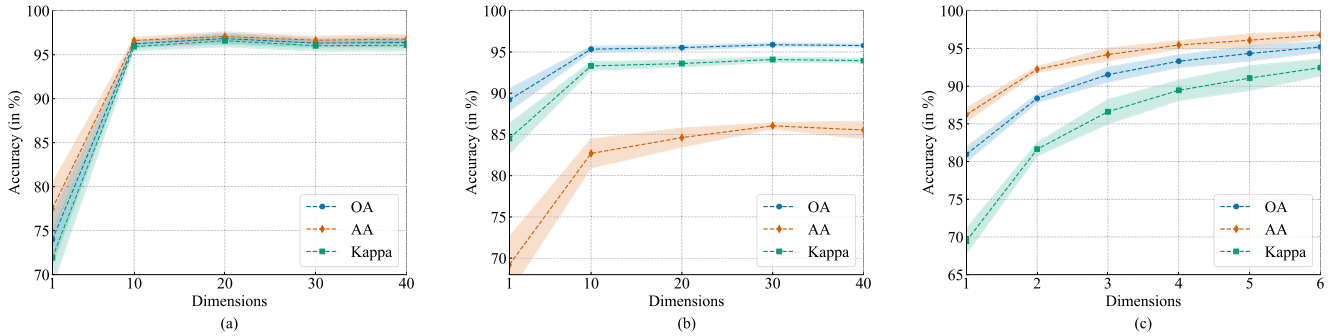


Fig. 8. Impact of different dimensions on accuracy in (a) Houston 2013 dataset, (b) Augsburg dataset, and (c) GRSS-DFC-2007 dataset.

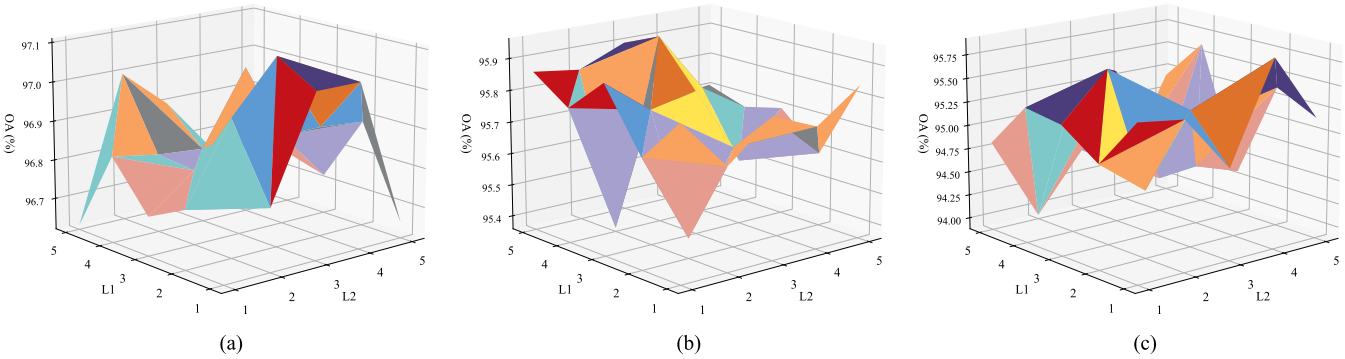


Fig. 9. OA of different models with different layers for LatSMB and LonSMB. (a) Houston 2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

2013 dataset, the best performance is observed at 20 dimensions, while in the Augsburg dataset, the optimal performance occurs at 30 dimensions. For the GRSS-DFC-2007 dataset, which consists of only six spectral bands, we experiment with dimensions from 1 to 6. The results indicate that classification accuracy steadily increases with more dimensions. Therefore, we select 20, 30, and 6 dimensions for the Houston 2013, Augsburg, and GRSS-DFC-2007 datasets, respectively.

3) Number of Layers for LatSMB and LonSMB: Fig. 9 illustrates the impact of varying numbers of layers, L_1 and L_2, in LatSMB and LonSMB on OA. The experiment combines different values of L_1 and L_2, both ranging from 1 to 5, resulting in 25 configurations tested for each dataset. The results indicate fluctuating performance across all three datasets. In the Houston 2013 dataset, the maximum OA is 97.10%, while the minimum is 96.63%, yielding a difference of 0.47%. When both L_1 and L_2 are set to 1, the OA reaches 96.98%, only 0.12% below the maximum. In the Augsburg dataset, the maximum OA is 95.95%, and the minimum is 95.37%, with a difference of 0.58%. For L_1 and L_2 set to 1, OA is 95.84% and 0.11% below the maximum. In the GRSS-DFC-2007 dataset, the maximum OA is 95.88%, and the minimum is 93.92%, resulting in a difference of 1.96%. When L_1 and L_2 are set to 1, the OA is 95.51% and 0.37% below the maximum. As observed across all three datasets, lower depths are sufficient to achieve competitive classification performance while maintaining minimal Flop consumption. Conversely, deeper depths provide limited accuracy improvements but linearly increase computational overhead. By trading off classification performance and computational complexity, the proposed Slice Mamba is constructed as a lightweight architecture, with both L_1 and L_2 set to 1.

D. Experimental Comparison With Competitive Approaches

We select several advanced algorithms in multisource fusion classification from recent years for comparison with the

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TABLE II

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACIES OF EACH CLASSES FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	AMSSE-Net	Coupled CNNs	CALC	LSAF	HCT	ExViT	NCGLF^2	FDNet	Slice Mamba
Health grass	94.55(4.05)	97.11(1.24)	93.15(4.15)	92.80(3.70)	94.56(2.79)	94.90(3.69)	94.68(1.18)	93.61(5.38)	**97.61(0.51)**
Stressed grass	95.80(5.45)	98.09(1.60)	97.89(1.52)	94.40(4.60)	96.61(3.45)	97.06(3.01)	96.08(2.53)	97.94(1.31)	**98.92(1.08)**
Synthetic grass	99.20(1.71)	98.02(1.42)	98.75(0.85)	**99.69(0.42)**	99.63(0.36)	98.35(1.18)	98.70(1.12)	99.06(1.75)	99.20(0.20)
Trees	97.51(2.32)	**98.62(0.69)**	96.00(2.59)	76.42(4.56)	98.05(1.62)	96.75(2.60)	95.69(2.42)	98.27(2.63)	98.22(2.02)
Soil	98.75(1.83)	98.74(1.04)	99.77(0.29)	99.85(0.18)	**99.93(0.19)**	99.82(0.27)	99.88(0.13)	99.65(0.47)	99.89(0.29)
Water	95.57(2.15)	**98.33(0.93)**	97.34(3.90)	97.28(2.70)	92.89(4.57)	98.03(1.39)	97.68(0.77)	97.75(1.11)	98.26(1.00)
Residential	90.50(5.86)	95.40(0.81)	**95.41(1.97)**	71.05(5.51)	86.05(4.12)	94.92(1.65)	94.00(3.67)	91.14(4.54)	95.10(2.31)
Commercial	84.88(7.65)	90.80(3.86)	93.69(1.71)	86.20(5.37)	86.05(6.48)	87.91(3.89)	94.07(4.01)	84.80(6.89)	**94.45(3.34)**
Road	69.61(12.28)	**89.22(2.17)**	87.73(3.35)	82.60(7.84)	78.78(6.86)	86.45(3.72)	84.55(4.69)	77.11(5.45)	85.41(4.36)
High way	91.25(5.89)	95.79(1.84)	95.00(3.45)	89.45(4.57)	91.22(6.16)	92.86(2.11)	99.41(0.98)	94.27(3.69)	**99.62(0.53)**
Rail way	94.90(3.44)	94.21(1.40)	95.79(4.40)	93.14(3.85)	96.20(3.75)	93.94(3.77)	94.75(3.28)	96.03(1.77)	**98.11(1.25)**
Parking_lot_1	90.26(4.61)	91.91(1.82)	93.90(2.38)	82.99(6.80)	91.51(7.21)	90.83(6.16)	94.16(1.81)	90.31(5.43)	**97.95(1.60)**
Parking_lot_2	84.61(8.05)	95.12(0.39)	96.33(2.12)	93.94(3.26)	**96.66(2.69)**	95.46(0.20)	91.62(2.61)	92.12(2.74)	93.06(0.88)
Tennis court	99.58(0.85)	99.93(0.16)	**100.00(0.00)**	99.46(0.80)	99.88(0.37)	99.65(0.51)	**100.00(0.00)**	99.69(0.50)	99.96(0.10)
Running track	99.16(1.29)	99.70(0.37)	**100.00(0.00)**	99.97(0.06)	99.72(0.37)	99.88(0.28)	99.95(0.17)	99.75(0.43)	**100.00(0.00)**
OA	91.69(2.95)	95.72(0.36)	95.47(0.90)	88.80(1.14)	93.64(1.12)	94.36(0.99)	95.24(0.63)	93.25(0.87)	**96.82(0.51)**
AA	92.41(2.65)	96.23(0.28)	96.05(0.74)	90.62(1.04)	94.61(0.91)	95.12(0.89)	95.68(0.56)	94.10(0.66)	**97.05(0.40)**
Kappa×100	91.02(3.19)	95.38(0.39)	95.10(0.98)	87.89(1.23)	93.12(1.20)	93.90(1.06)	94.85(0.68)	92.70(0.94)	**96.56(0.56)**

TABLE III

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACIES OF EACH CLASSES FOR THE AUGSBURG DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	AMSSE-Net	Coupled CNNs	CALC	LSAF	HCT	ExViT	NCGLF^2	FDNet	Slice Mamba
Forest	98.61(0.51)	98.11(0.51)	97.90(0.41)	**98.88(0.17)**	97.33(0.64)	97.45(1.03)	98.02(0.64)	98.86(0.46)	98.31(0.34)
Residential-Area	92.94(1.53)	95.93(0.61)	**97.01(0.63)**	94.90(0.54)	94.11(2.09)	95.42(0.98)	96.13(1.01)	93.17(1.05)	96.29(0.53)
Industrial-Area	79.39(4.00)	87.54(1.73)	**90.85(3.45)**	87.81(0.95)	86.36(1.54)	82.34(5.06)	84.54(4.69)	84.51(3.21)	86.64(2.57)
Low-Plants	**99.23(0.15)**	99.07(0.13)	98.38(0.41)	99.29(0.06)	99.08(0.16)	98.63(0.38)	98.63(0.33)	98.86(0.24)	98.87(0.15)
Allotment	92.67(3.38)	91.01(2.98)	73.97(7.23)	**95.91(0.58)**	94.86(3.34)	88.74(7.77)	92.24(1.99)	92.69(1.05)	94.72(1.33)
Commercial-Area	11.44(4.84)	12.21(5.27)	0.88(2.77)	16.90(1.72)	35.73(4.35)	14.77(4.57)	34.51(5.16)	17.67(8.17)	**40.87(11.28)**
Water	87.54(1.89)	90.70(1.79)	84.79(3.32)	**92.37(0.90)**	90.66(1.81)	84.01(3.41)	90.11(1.80)	88.18(1.33)	90.56(2.99)
OA	93.55(0.75)	95.05(0.26)	94.93(0.18)	95.02(0.25)	94.67(0.82)	94.26(0.46)	95.30(0.60)	93.96(0.35)	**95.75(0.29)**
AA	80.26(1.56)	82.08(0.69)	77.68(1.21)	83.72(0.45)	85.45(0.74)	80.19(1.56)	84.88(1.13)	81.99(0.90)	**86.61(1.51)**
Kappa×100	90.82(1.04)	92.92(0.37)	92.71(0.25)	92.89(0.35)	92.41(1.13)	91.79(0.65)	93.25(0.84)	91.41(0.48)	**93.92(0.40)**

TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, KAPPA, AND THE ACCURACIES OF EACH CLASSES FOR THE GRSS-DFC-2007 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	AMSSE-Net	Coupled CNNs	CALC	LSAF	HCT	ExViT	NCGLF^2	FDNet	Slice Mamba
City Center	85.26(2.17)	92.77(1.16)	96.77(1.61)	86.91(2.21)	92.30(1.94)	92.50(1.13)	96.07(0.77)	79.26(1.73)	**96.91(0.91)**
Residential Areas	88.78(1.00)	92.86(0.60)	**96.36(1.22)**	87.29(2.22)	91.92(1.95)	91.38(0.78)	95.76(0.48)	80.73(2.03)	96.15(0.65)
Sparse Building	73.54(2.78)	84.58(1.84)	89.61(3.36)	66.46(6.76)	88.63(2.68)	81.68(1.13)	88.91(2.74)	60.40(2.63)	**91.29(2.67)**
Water	94.54(2.64)	98.04(1.18)	99.43(1.09)	96.59(1.55)	92.41(3.78)	98.46(1.03)	99.21(1.13)	94.44(2.28)	**99.58(0.52)**
Vegetation	96.54(5.64)	**100.00(0.00)**	99.62(0.55)	94.17(4.93)	99.68(0.60)	99.82(0.34)	**100.00(0.00)**	98.71(1.73)	99.90(0.28)
OA	84.30(1.07)	90.90(0.40)	94.83(0.63)	82.16(1.29)	91.26(0.56)	89.37(0.35)	94.19(0.70)	75.66(0.78)	**95.19(0.98)**
AA	87.73(1.08)	93.65(0.38)	96.39(0.75)	86.28(2.34)	92.99(0.82)	92.77(0.36)	95.99(0.62)	82.71(0.98)	**96.77(0.72)**
Kappa×100	74.99(1.76)	85.65(0.66)	91.85(1.03)	71.54(2.41)	86.28(0.85)	83.25(0.54)	90.86(1.12)	61.27(1.23)	**92.44(1.55)**

method proposed in this article. To ensure fairness, the experiments use the parameter settings provided in the original papers. The algorithms chosen for comparison include: coupled CNNs [18], the coupled adversarial learning-based classification method (CALC) [44], the linear self-attention fusion method (LSAF) [45], the hierarchical CNN and Transformer (HCT) [28], the extended vision Transformer (ExViT) [46], the network combining global and local features for fusion (NCGLF^2) [5], the frequency-domain-based network (FDNet) [27], and the adaptive multiscale spatial–spectral enhancement network (AMSSE-Net) [47].

Tables II–IV present the performance comparison of various classification methods across three datasets, using metrics such as OA, AA, Kappa, and the classification accuracy for each category. Figs. 10–12 illustrates the visualization maps of the algorithms on these datasets. Fig. 13 provides

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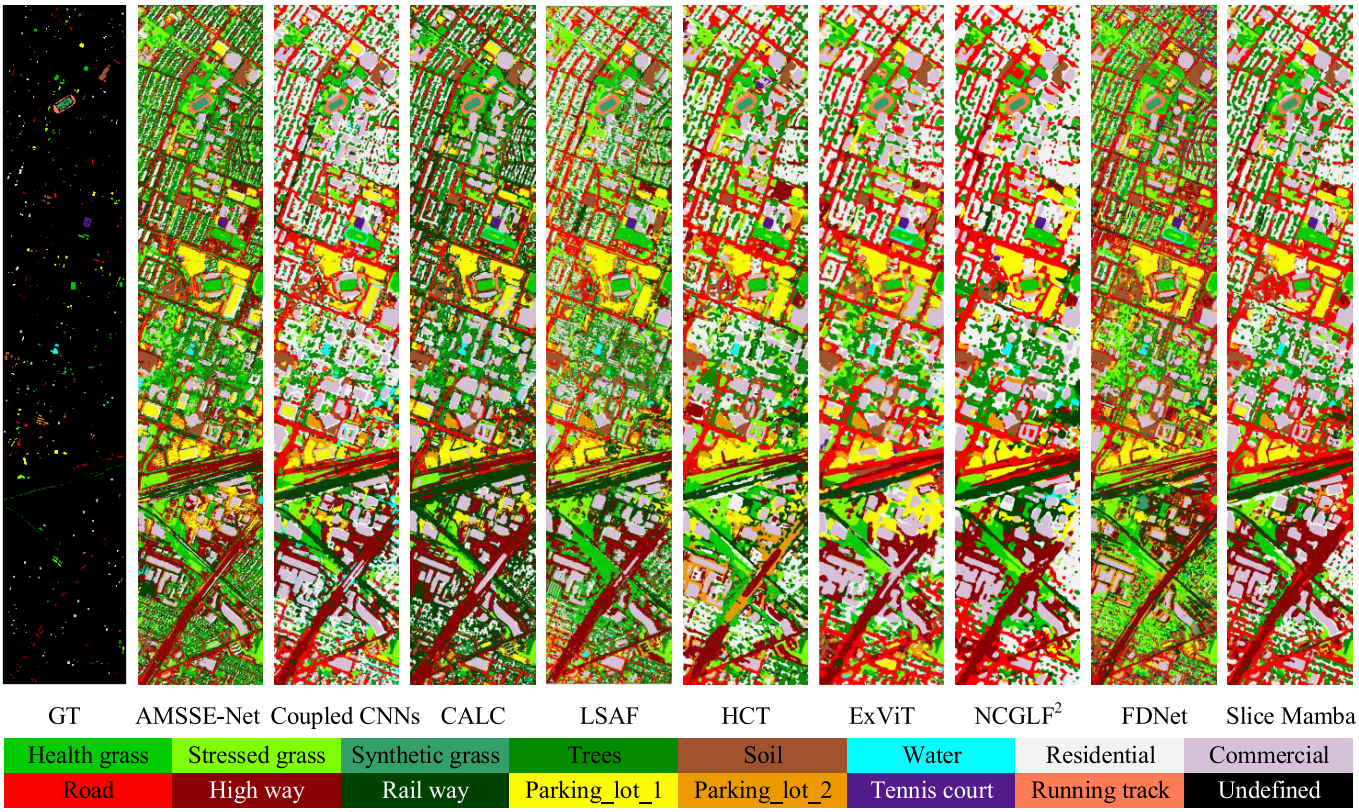


Fig. 10. Classification maps obtained by the compared methods on the Houston 2013 dataset.

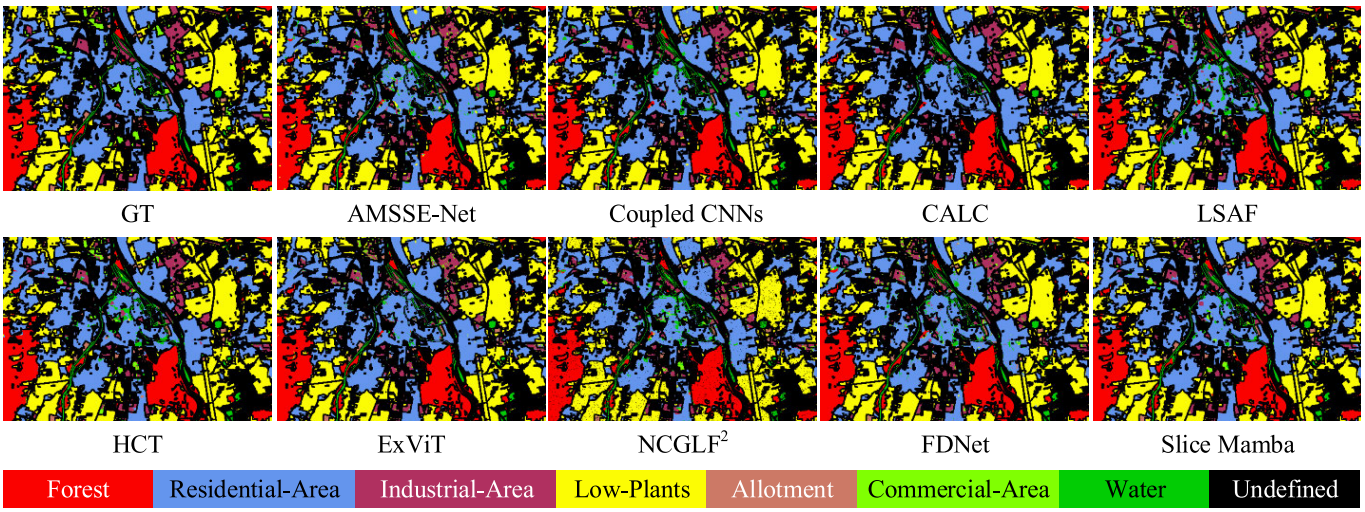


Fig. 11. Classification maps obtained by the compared methods on the Augsburg dataset.

the corresponding confusion matrices for each dataset. In all datasets, Slice Mamba outperforms other comparative algorithms in terms of OA, AA, and Kappa accuracy. This demonstrates that Slice Mamba not only exhibits competitive performance in the fusion classification of HSI and LiDAR data but also maintains excellent performance when extended to the fusion of MSI and SAR data. By utilizing the feature extraction modules of LatSMB and LonSMB, along with the CIF module, our algorithm successfully generalizes the features learned from HSI data to MSI images.

1) Houston 2013 Dataset: The Houston 2013 dataset contains regions covered by clouds, which increases the complexity of the multisource data fusion classification task. Table II shows that the Slice Mamba method demonstrates outstanding overall performance, achieving the highest OA of 96.82%, AA of 97.05%, and Kappa coefficient of 96.56%. Compared with the second-best algorithm, coupled CNNs, Slice Mamba improves OA by 1.10%, AA by 0.82%, and Kappa by 1.18%. In addition, Slice Mamba achieves the best classification accuracy in categories such as health grass,

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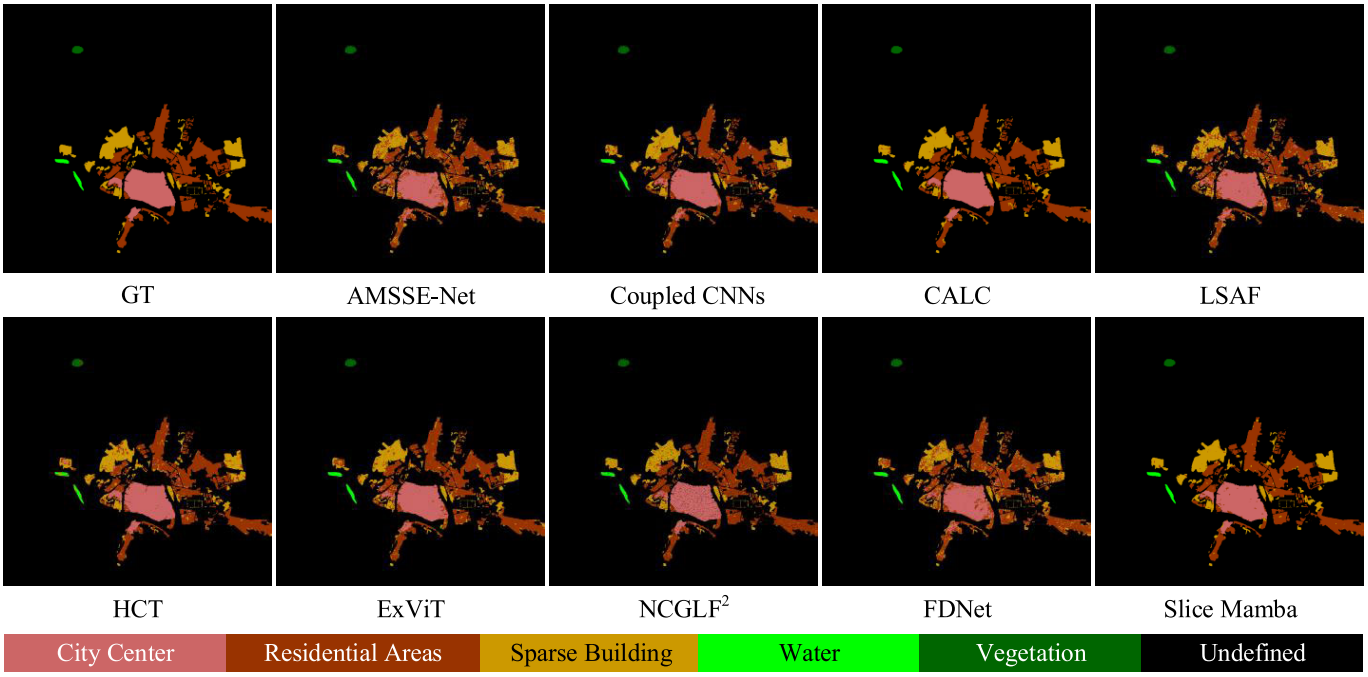


Fig. 12. Classification maps obtained by the compared methods on the GRSS-DFC-2007 dataset.

stressed grass, residential, highway, railway, parking_lot_1, and running track, while its performance in other categories is comparable with the best results. As illustrated in Fig. 10, our method produces smoother classification results in cloud-covered areas, such as highway and residential, outperforming other algorithms. From Fig. 13, the matrix highlights the classification accuracy across 15 land cover classes. High classification performance is observed for most classes, as indicated by the predominantly blue diagonal. These phenomena indicate that Slice Mamba offers greater robustness and competitiveness in handling challenging environments compared with other methods.

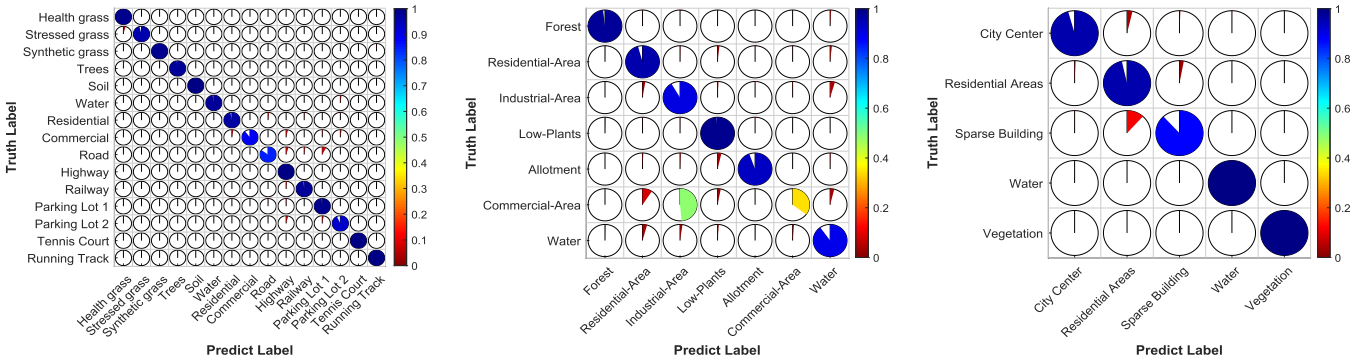
2) Augsburg Dataset: The Augsburg dataset exhibits an imbalanced distribution of samples, with the Commercial-Area region scattered across multiple areas of the map, posing greater challenges for the algorithms. Table III presents the classification results for this dataset, where Slice Mamba achieves the best performance in terms of OA, AA, and Kappa. Notably, in the commercial-area region, Slice Mamba outperforms the second-best algorithm, with an OA improvement of 6.36%. CALC employs a generative adversarial network (GAN)-based unsupervised learning approach, making it more sensitive to imbalanced samples. As a result, it performs well in categories with larger sample sizes, such as industrial-area, but struggles in those with fewer samples. LSAF leverages a linear self-attention module for feature extraction, demonstrating strong performance in recognizing large regions. In contrast, Slice Mamba shows the superior robustness when handling imbalanced samples, particularly excelling at identifying small and scattered regions. As illustrated in Fig. 13, a strong classification performance is evident for the "low-plants" and "forest" classes. However, notable misclassifications occur between the "commercial-area" and "industrial-area" classes, due to the inherent spatial and spectral similarities between these two land cover types. The performance demonstrates the efficacy of the proposed Slice Mamba while highlighting specific challenges in differentiating similar land cover types.

3) GRSS-DFC-2007: The GRSS-DFC-2007 dataset is primarily used to evaluate the adaptability of algorithms to MSI data. Table IV shows that Slice Mamba achieves superior performance in terms of OA and obtains the highest classification accuracy in categories such as city center, sparse building, and vegetation. Moreover, its performance in other categories differs by not more than 0.3% from the best algorithm. As illustrated in Fig. 12, Slice Mamba produces smoother classification results compared with other algorithms, with significantly fewer noise artifacts in the images. This demonstrates that Slice Mamba exhibits impressive performance in multisource data fusion and adaptability, further validating its effectiveness in complex tasks. In addition, the "water" and "vegetation" classes demonstrate near-perfect classification accuracy, whereas "sparse building" exhibits noticeable misclassifications, as evidenced by nonblue off-diagonal entries in Fig. 13.

E. Ablation Study

1) Ablation Analysis of Different Modules in Slice Mamba: To validate the effectiveness of each module in our model, we conduct an ablation study on three datasets: Houston 2013, Augsburg, and GRSS-DFC-2007, the experimental results are presented in Table V. When the MIF module is not used, the OA accuracy of the three datasets decreases by 0.74%, 0.49%, and 0.46%, respectively. The MIF module integrates multiscale information, effectively extracting contour features. Due to the high number of classes and rich edge regions in the Houston 2013 dataset, the extraction of multiscale

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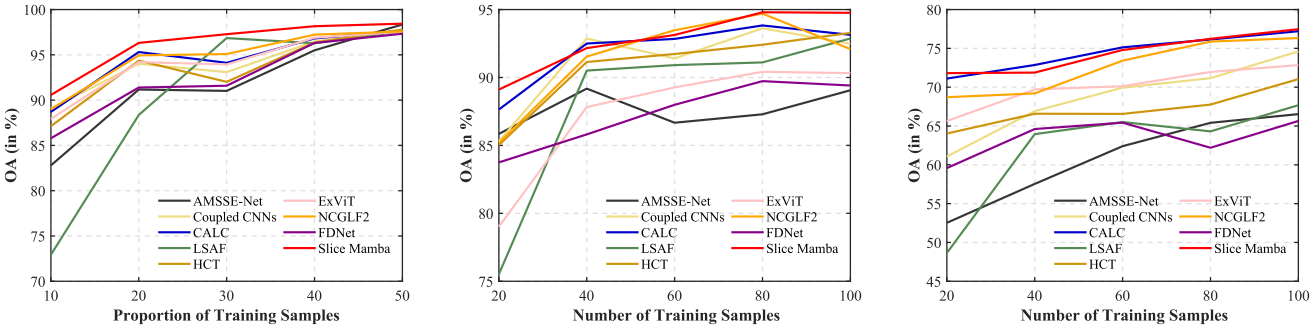
(a) (b) (c)

Fig. 13. Confusion matrix obtained for the proposed Slice Mamba with the three multisource datasets. (a) Houston 2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

TABLE V

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE HOUSTON2013, AUGSBURG, AND GRSS-DFC-2007 DATASETS

MIF	SFE	CIF	LatSMB	LonSMB	Houston 2013			Augsburg			GRSS-DFC-2007		
					OA	AA	Kappa×100	OA	AA	Kappa×100	OA	AA	Kappa×100
×	✓	✓	✓	✓	96.08	95.75	94.92	95.26	85.90	93.23	94.73	96.45	92.16
✓	×	✓	✓	✓	96.04	96.51	95.72	95.23	84.05	93.15	94.35	95.15	89.51
✓	✓	×	✓	✓	95.44	95.70	95.07	94.83	84.05	92.87	93.35	95.15	89.51
✓	✓	✓	×	✓	95.62	95.84	95.26	95.07	84.66	92.94	94.22	94.55	90.84
✓	✓	✓	✓	×	95.98	96.56	95.66	95.42	86.27	93.44	94.98	96.75	92.12
✓	✓	✓	✓	✓	96.82	97.05	96.56	95.75	86.61	93.92	95.19	96.77	92.44



(a) (b) (c)

Fig. 14. Classification performance of the compared methods with different numbers of training data on three datasets. (a) Houston 2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

information is more sensitive, resulting in a more significant accuracy decline. When the SPE module is removed, the OA accuracy of the three datasets decreases by 0.78%, 0.52%, and 0.84%, respectively. The SFE module extracts features through LatSMB and LonSMB, preparing for data fusion. The results obtained after removing the SFE module further demonstrate its necessity prior to fusion. In the absence of the CIF module, when directly fusing the HSI and LiDAR features output from the SFE module, we observe that the OA accuracy decreases by 1.36%, 0.92%, and 1.84% across the three datasets. The CIF module aims to fuse heterogeneous features and forms a complementary structure with the SFE module, allowing for comprehensive feature extraction. Given that the GRSS-DFC-2007 dataset comprises MSI and SAR data, simple direct fusion fails to adequately integrate the heterogeneous features, making performance more sensitive on this dataset. The experimental results indicate that the CIF module effectively fuses heterogeneous features, thereby enhancing the algorithm performance.

2) Ablation Analysis of Different Slices in Slice Mamba: To further investigate the effectiveness of the combination of LatSMB and LonSMB, we design ablation experiments by excluding LatSMB and LonSMB, respectively. When LatSMB is not used, the OA accuracy decreases by 1.20%, 0.68%, and 0.97% across the three datasets. In contrast, when LonSMB is omitted, the OA accuracy decreases by 0.84%, 0.33%, and 0.21% for the three datasets. The results, in conjunction with the experiments excluding the CIF module, indicate that the combined use of LatSMB and LonSMB significantly enhances the performance of the algorithm.

F. Robustness Assessment

To assess the robustness of the proposed Slice Mamba, extensive experiments are performed by varying the number of

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TABLE VI
COMPUTATIONAL PARAMETER ANALYSIS OF VARIOUS COMPARISON APPROACHES IN TERMS OF PARAMETERS AND FLOPS ON THE HOUSTON 2013 DATASET

Metrics	AMSSE-Net	Coupled CNNs	CALC	LSAF	HCT	ExViT	NCGLF^2	FDNet	Slice Mamba
Params(M)	0.75	0.10	0.30	0.22	0.43	0.23	0.44	0.40	0.28
FLOPs(G)	0.91	0.15	2.18	7.52	0.51	2.94	4.36	0.90	2.71

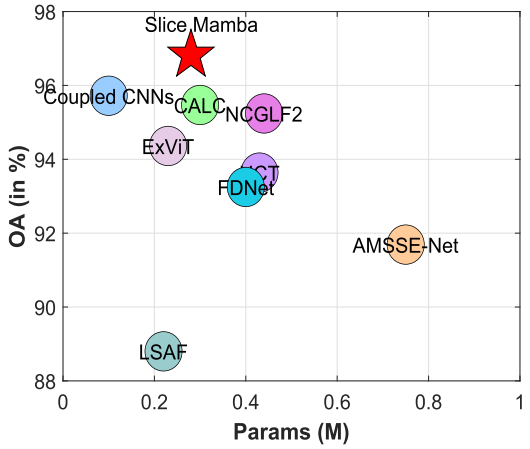


Fig. 15. Computational analysis of various comparison approaches in terms of accuracy (OA) and parameters on the Houston 2013 dataset.

training samples. Fig. 14 illustrates the performance variations on the three datasets, with the proposed Slice Mamba prominently highlighted by red curves. The selected numbers for the Houston 2013 datasets range from 10 to 50 in increments of 10, while for the Augsburg and GRSS-DFC-2007 datasets, the numbers are set at 20, 40, up to 100 in increments of 20. From the results, the proposed Slice Mamba exhibits a stable and significant improvement in accuracy as the number of training samples grows, outperforming all competing methods consistently across different training sample ranges. With a smaller number of training samples (e.g., 20 or fewer), the performance gap between Slice Mamba and other methods is particularly pronounced. This highlights the robustness of the proposed method in practical scenarios where training data are limited. In summary, the number of training samples has a significant impact on model performance, and Slice Mamba demonstrates robust performance across varying sample sizes.

G. Analysis of Computational Complexity

This section measures the computational efficiency of various algorithms by analyzing their parameters and floating point operations (FLOPs) on the Houston 2013 dataset, with specific results presented in Table VI. Coupled CNNs and ExViT employ a weight-sharing strategy, resulting in a reduced number of parameters. The LSAF algorithm utilizes a linear self-attention mechanism, achieving a similarly low parameter count. In comparison with Transformer-based methods, the proposed algorithm relies on linear complexity, demonstrating a clear advantage in parameter count due to the unique linear complexity of Mamba. As illustrated in Fig. 15, the proposed Slice Mamba, represented by the red star, achieves the highest classification performance while maintaining a lightweight architecture with relatively fewer model parameters. While CNN-based methods like coupled CNNs exhibit lower computational burdens, their ability to capture global spatial–spectral features is limited by the constraints of local receptive fields, restricting further performance improvement. These results demonstrate that Slice Mamba achieves an optimal tradeoff between computational efficiency and classification effectiveness, highlighting significant potentiality and viability for practical applications.

IV. CONCLUSION

This article proposes a multisource RS data fusion classification method based on the Mamba architecture, termed Slice Mamba. The method first utilizes the MIF module to effectively integrate multiscale information, extracting shallow features. The LatSS and LonSS scanning mechanisms are then employed to enhance information interaction between pixels, forming the LatSMB and LonSMB modules, which together comprise the SFE module for capturing spatial structure and spectral information from multisource RS data. Subsequently, the CIF module establishes a complementary scanning structure to fully integrate heterogeneous features, ultimately accomplishing the fusion classification task. Extensive experiments demonstrate that our algorithm performs exceptionally well in the fusion classification of HSI and LiDAR data and shows good generalization capability in the fusion classification of MSI and SAR data. However, we recognize that the multidirectional scanning mechanism increases computational complexity. The future research focuses on reducing the computational complexity of the model and further improving computational efficiency to facilitate the broader application of the Mamba model in hyperspectral scenarios.

REFERENCES

[1] X. Liao, B. Tu, J. Li, and A. Plaza, "Class-wise graph embedding-based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522813.
[2] A. Vali, S. Comai, and M. Matteucci, "Deep learning for land use and land cover classification based on hyperspectral and multispectral Earth observation data: A review," *Remote Sens.*, vol. 12, no. 15, p. 2495, Aug. 2020.
[3] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2023.
[4] J. Geng, X. Deng, X. Ma, and W. Jiang, "Transfer learning for SAR image classification via deep joint distribution adaptation networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5377–5392, Aug. 2020.
[5] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192.
[6] S. Feng, H. Zhang, B. Xi, C. Zhao, Y. Li, and J. Chanussot, "Cross-domain few-shot learning based on decoupled knowledge distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534414.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:50:46 UTC from IEEE Xplore. Restrictions apply.

[7] B. Qin, S. Feng, C. Zhao, B. Xi, W. Li, and R. Tao, "FDGNet: Frequency disentanglement and data geometry for domain generalization in cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, Aug. 2024.

[8] Z. Qiu, J. Xu, J. Peng, and W. Sun, "Domain fusion contrastive learning for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501812.

[9] W. Yu, L. Gao, H. Huang, Y. Shen, and G. Shen, "HI2D2FNet: Hyperspectral intrinsic image decomposition guided data fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521715.

[10] K. Li, D. Wang, X. Wang, G. Liu, Z. Wu, and Q. Wang, "Mixing self-attention and convolution: A unified framework for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5523216.

[11] J. Wang, W. Li, Y. Wang, R. Tao, and Q. Du, "Representation-enhanced status replay network for multisource remote-sensing image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15346–15358, Nov. 2024.

[12] X. Geng et al., "Multisource joint representation learning fusion classification for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4406414.

[13] Y. Gao, M. Zhang, W. Li, X. Song, X. Jiang, and Y. Ma, "Adversarial complementary learning for multisource remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505613.

[14] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 8057–8070, Oct. 2023.

[15] Y. Liu, Z. Ye, Y. Xi, H. Liu, W. Li, and L. Bai, "Multiscale and multidirection feature extraction network for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9961–9973, 2024.

[16] B. Yang, Y. Hu, X. Liu, and J. Li, "CEFusion: An infrared and visible image fusion network based on cross-modal multi-granularity information interaction and edge guidance," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 17794–17809, Nov. 2024.

[17] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[18] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[19] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[20] B. Yang, X. Wang, Y. Xing, C. Cheng, W. Jiang, and Q. Feng, "Modality fusion vision transformer for hyperspectral and LiDAR data collaborative classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17052–17065, 2024.

[21] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[22] Y. Kong, Y. Cheng, Y. Chen, and X. Wang, "Joint classification of hyperspectral image and LiDAR data based on spectral prompt tuning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521312.

[23] S. K. Roy, A. Sukul, A. Jamali, J. M. Haut, and P. Ghamisi, "Cross hyperspectral and LiDAR attention transformer: An extended self-attention for land use and land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512815.

[24] J. X. Yang, J. Zhou, J. Wang, H. Tian, and A. W.-C. Liew, "LiDAR-guided cross-attention fusion for hyperspectral band selection and image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515815.

[25] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2022.

[26] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[27] K. Ni, D. Wang, G. Zhao, Z. Zheng, and P. Wang, "Hyperspectral and LiDAR classification via frequency domain-based network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525117.

[28] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[29] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[30] J. Ma, F. Li, and B. Wang, "U-mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, *arXiv:2401.04722*.

[31] T. Y. Tsai, L. Lin, S. Hu, M.-C. Chang, H. Zhu, and X. Wang, "UU-mamba: Uncertainty-aware U-mamba for cardiac image segmentation," in *Proc. IEEE 7th Int. Conf. Multimedia Inf. Process. Retr. (MIPR)*, Aug. 2024, pp. 267–273.

[32] G. Li, Q. Huang, W. Wang, and L. Liu, "Selective and multi-scale fusion mamba for medical image segmentation," *Expert Syst. Appl.*, vol. 261, Feb. 2025, Art. no. 125518.

[33] C. Ma and Z. Wang, "Semi-Mamba-UNet: Pixel-level contrastive and cross-supervised visual mamba-based UNet for semi-supervised medical image segmentation," *Knowl.-Based Syst.*, vol. 300, Sep. 2024, Art. no. 112203.

[34] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[35] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-mamba: Interval group spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.

[36] G. Zhao, H. Wu, D. Luo, X. Ou, and Y. Zhang, "Spatial–spectral interaction super-resolution CNN–Mamba network for fusion of satellite hyperspectral and multispectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18489–18501, 2024.

[37] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[38] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[39] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[40] L. Huang, Y. Chen, and X. He, "Spectral–spatial mamba for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 13, p. 2449, Jul. 2024.

[41] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[42] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[43] F. Pacifici, F. Del Frate, W. J. Emery, P. Gamba, and J. Chanussot, "Urban mapping using coarse SAR and optical data: Outcome of the 2007 GRSS data fusion contest," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 3, pp. 331–335, Jul. 2008.

[44] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[45] M. Feng, F. Gao, J. Fang, and J. Dong, "Hyperspectral and LiDAR data classification based on linear self-attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jun. 2021, pp. 2401–2404.

[46] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[47] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

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